Collaborative modeling for groundwater sustainability evaluation in coastal aquifers using simulation optimization

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Highlights

1. Participatory modeling (PM) simulation optimization (SO) for groundwater sustainability
2. Unlike traditional SO that treats participation as a separate or post hoc step
3. Direct embedding of societal values and policy constraints in core formulation of SO
4. Method makes institutional reality like groundwater availability vs. accessibility testable
5. PM avoids technically optimal solutions that are institutionally infeasible

Abstract

Sustainable groundwater management of coastal aquifers is challenging due to saltwater intrusion, climate change, and competing societal preferences. This demand approaches that are both scientifically rigorous and socially legitimate. This study introduces a stakeholder-informed simulation–optimization framework, which directly integrates collaborative modeling with stakeholders into the core formulation of the simulation optimization problem, unlike traditional simulation optimization approaches that treats participation as a separate or post hoc step. Applied to Pearl Harbor aquifer in Hawai'i, the modeling team and water managers co-designed the simulation optimization decision variables, objectives, constraints, and scenario analysis, grounding the simulation optimization in management reality. This integration embedded management settings and goals, societal values, and ecological thresholds in the simulation optimization to co-define feasible solutions, thereby enhancing model credibility, legitimacy, and salience. Results show that projected climate change for a dry scenario with sea-level rise could cause a 46% collapse in sustainable withdrawal by mid-century due to saltwater intrusion. The stakeholder-informed simulation optimization mitigates these impacts, improving freshwater availability by up to 30% compared to non-optimized approaches. Combining optimization with nature-based solutions like watershed restoration through forest protection and corridor development provides synergistic benefits, adding up to 10% more sustainable yield. The framework also quantifies key policy trade-offs, such as the balance between maximizing withdrawal and maintaining spring discharge for beneficial use including ecological and cultural uses. By operationalizing stakeholder knowledge within the simulation optimization framework, this work offers a transferable collaborative modeling for advancing groundwater sustainability in coastal regions worldwide.

Keywords

simulation optimization, sustainable groundwater management, stakeholder engagement, participatory modeling, coastal aquifers, climate change, nature-based solutions, saltwater intrusion

Plain language summary

Coastal groundwater is threatened by saltwater intrusion and climate change. Purely technical solutions often fail to address real-world management needs. To find better solutions, we developed a method that integrates the knowledge of water managers directly into the core of a computer simulation model. By co-designing the model with stakeholders in Hawai'i, we created a tool that finds realistic pumping strategies to mitigate climate impacts, protect ecological flows, evaluate nature-bases solutions, and balance competing needs. A collaborative approach between modeling team and stakeholders produces more trusted and effective management plans for sustainable groundwater management.

1. Introduction

Sustainable groundwater management requires tools that are not only scientifically rigorous but also socially legitimate (Alegria et al., 2024; Chen et al., 2025; Cuthbert et al., 2023; A. S. Elshall, Arik, et al., 2020; Gleeson et al., 2020; Karandish et al., 2025; Perrone et al., 2023; Rohde et al., 2024; Sadeghi-Jahani et al., 2024; Wang et al., 2024). These are tools that translate ecological thresholds, policy mandates, and community values into defensible decisions under uncertainty (Perrone et al., 2023; Pierce et al., 2013). This is particularly true for coastal aquifers, which are critical water sources for rapidly growing populations yet face intensifying pressures from saltwater intrusion, sea-level rise, shifting recharge, and competing societal preferences (Elshall, et al., 2020; Michael et al., 2017). Thus, in groundwater policy (e.g., EU Water Framework Directive; California's SGMA; Hawai'i Water Code), stakeholder involvement is not merely advisable but often required (Elshall et al., 2022), both to raise awareness and to ensure that management plans reflect diverse public interests (Perrone et al., 2023). Accordingly, participatory and collaborative modelling (hereafter "participatory modelling") between modeling team and stakeholders has become central to environmental decision-support because it strengthens the *credibility*, *legitimacy*, and *salience* of analyses(Bandari et al., 2023; Gray et al., 2018; Hedelin et al., 2021; Jordan et al., 2018; Karki et al., 2025; Langsdale et al., 2013; Salas & Pennington, 2024; Shuler & Mariner, 2020; Sterling et al., 2019; Voinov et al., 2016, 2018). When designed well, participatory processes surface local and institutional knowledge, build trust, improve assumptions, and help reconcile conflicting perspectives in complex socio-hydrological systems in groundwater management (Hinton et al., 2025; Iwanaga et al., 2020; Kniffin et al., 2020; Manda & Klein, 2014; Martin et al., 2021; Mussehl et al., 2023; Perdikaki et al., 2022; Perrone et al., 2023; Zare et al., 2021).

In parallel, simulation–optimization is widely used in water resources and environmental management (Bista et al., 2025; Duraisekaran et al., 2025; J. Li et al., 2023; Yang et al., 2024). Coupling numerical simulators with optimization algorithms, simulation optimization identifies feasible solutions and tradeoffs given the management objectives. Simulation optimization In groundwater management (Gorelick et al., 2015) has been widely used in many groundwater management applications. This includes groundwater contamination remediation and source identification (Chang et al., 2024; Elshall, Ye, et al., 2020; Lee et al., 2024), monitoring network design (Farshad et al., 2025; Song et al., 2019), hydroeconomic modeling(G. Li et al., 2023; Mulligan et al., 2014), management of groundwater dependent ecosystems (Han et al., 2023), designing of nature-based infrastructure (Torres et al., 2021; Zhang & Chui, 2024), optimizing irrigation and farming processes (Fowler et al., 2015; Hou et al., 2025), coastal aquifer management (Yu et al., 2023; Zang et al., 2023), and groundwater sustainability evaluation(Farhadi et al., 2016; Hesamfar et al., 2023; Ma et al., 2024; Mishra

et al., 2025; Roy & Datta, 2025; Yazdian et al., 2021; Zhao et al., 2024). This study advances participation-based groundwater management and simulation optimization-based management by formally integrating participatory modeling with simulation-optimization to add credibility, legitimacy, and saliency to the technical assessment.

While numerous variants of participation-based modeling have been used in environmental and water resources management (Basco-Carrera et al., 2017; Langsdale et al., 2013), a few studies have integrated these approaches within simulation optimization framework. For example, Raei et al. (2017) integrated a bioremediation simulation model with multi-objective optimization for optimal design of in situ groundwater bioremediation system, considering preferences of stakeholders. Similarly, Fowler et al. (2015) coupled the groundwater model MODFLW- Farm Process (MODFLOW-FMP2) with an optimization framework to identify trade-offs in crop selection under water-limited conditions. These two studies highlighted that participatory modeling combined with optimization is essential not only for dialogue but also for quantifying competing objectives and identifying management strategies acceptable to multiple stakeholders. Shen et al. (2023) developed a web-based participatory planning system with a multiobjective simulation optimization backend to assist expert and non-expert stakeholders in reaching a consensus on the optimal achievement of the watershed management goals. Other studies (Bremer et al., 2021; Burnett et al., 2020) focused on sustainable groundwater management. However, existing frameworks rarely provide a generalizable process that specifies where and how stakeholder inputs enter the objective space, how those inputs reshape trade-offs. This study is unique as it details the embedding of the stakeholders' knowledge in including stakeholder values and institutional constraints in the mathematical formulation of simulation optimization where decision variables, objectives, and constraints are set. A second contribution of this study is that it discusses how participatory modeling enhances credibility, legitimacy and saliency of technical solutions of simulation optimization. Methodologically, we provide guidelines for embedding participatory modelling inside simulation–optimization. Through a structured co-design process, we show the modeling team and water managers can jointly (i) select decision variables that reflect management reality such as different user groups, and which wells to optimize; (ii) co-define objectives such as maximizing withdrawal while minimizing salinization and drawdown and conserving ecological flows; (iii) explicit include ecological constraints and tradeoffs for terrestrial and marine ecosystems; and (iv) define and prioritize scenario analysis including linking climate change with respect to recharge and sea-level rise and nature-based solutions for watershed restoration and future development. In doing so, we make institutional realities such as the distinction between groundwater availability reflecting aquifer physical capacity and groundwater accessibility reflecting feasible withdrawal under policy (Asante-Sasu et al., 2025) explicit and testable

within the simulation optimization. In addition, in costal aquifers this is particularly critical where ecological performance constraints (e.g., spring protection and submarine groundwater discharge) and potable-water quality thresholds (e.g., chloride limits for potable versus irrigation supply) must be respected alongside water supply objective.

We demonstrate this stakeholder-driven method by developing and applying a simulation-optimization tool to evaluate groundwater sustainability of the Pearl Harbor (Puʻuloa) aquifer in Hawaiʻi, which isheavily utilized coastal aquifer supplying more than one-third of the state's population. This application showcases the framework's innovative capabilities for advancing management of coastal aquifers with ecological constraints. By operationalizing stakeholder knowledge within a rigorous quantitative framework, this work offers a transferable method for advancing groundwater sustainability in complex coastal regions worldwide. More generally, this replicable framework provides guidance for stakeholder-driven simulation optimization for environmental and water resource management, specifically for groundwater management.

The remainder of this paper details this process. Section 2 outlines the groundwater sustainability policy in Hawaiʻi and the technical components of our simulation-optimization method, which couples a 3D density-dependent groundwater model with the Hawaiʻi Water Budget Model (WBM) and a parallel-computing optimization algorithm. This purely technical model served as the starting point for the collaborative modeling process presented in Section 3. There, we describe the co-design process with water managers that transformed the model into a boundary object for collective reasoning about salinization risk, drawdown thresholds, ecological flows, societal preferences, and other policy-relevant trade-offs. Section 4 presents the results, demonstrating how this collaborative approach adds value and generates insights unattainable through a purely technical optimization. Finally, the Conclusions section synthesizes our key findings, discusses the broader implications of the co-design method, and underscores the framework's transferability for advancing sustainable groundwater management.

2. Case study

2.1 Groundwater sustainability policy in Hawai'i

Modern groundwater management in Hawaiʻi is continuously developing to better align with groundwater policy in the state. The state-mandated water code stipulates that the water regulators should estimate groundwater sustainable yield for all major aquifer systems in Hawaiʻi (HRS Chapter 174C, 1987). As such, sustainable yield evaluation is a major component of the Hawaiʻi Water Resources Protection Plan. The State Water Code (HRS Chapter 174C, 1987) defines sustainable yield as "the maximum rate at which water may be withdrawn from a source without impairing the utility or quality of the water source as

determined by the commission." The commission refers to the Commission on Water Resources Management (CWRM), which is the state water regulator. The definition of "without impairing the utility or quality of the water source" involves several aquifer performance and governance factors.

Sustainable yield has been widely calculated in Hawai'i using an analytical model known as the robust analytical groundwater flow and salinity transport model (RAM2, Liu & Dai, 2012; Liu, 2007). Rigorously derived from groundwater flow and transport equations, RAM2 is a simple mathematical tool to estimate aquifer yield in basal coastal aquifers by simulating the variation of the hydraulic head and the expansion of the transition zone under pumping stress. RAM2 is an easy-to-use model with only few parameters that can be applied satisfactorily with limited field monitoring data (Liu & Dai, 2012). However, the RAM2 model has two limitations. First, it estimates only safe yield since it does not account for induced recharge. Induced recharge is part of the groundwater captured caused by groundwater development. This includes public trust water uses such as spring discharge, recharge from surface water bodies, and increase in lateral inflow (in administratively defined but physically unbounded groundwater formations). Also, RAM2 lumps leakage into one term. Accordingly, RAM2 does not provide insights about separate leakage components such as submarine groundwater discharge, spring discharge, base-flow, evapotranspiration, drains, and decrease in lateral inflow. Quantifying different leakage components is important for managing groundwater dependent ecosystems, which is a stakeholder priority in Hawai'i (Adler et al., 2018). Second, RAM2 is a lumped model that does not provide insights about the spatial distribution of withdrawals with respect to the spatial distribution of hydraulic conductivity, dispersity, porosity, recharge, and other model parameters. For example, Oki & Meyer (2001) compare results obtained by a distributed numerical model and by RAM with field measurements and find that RAM underestimates water-level declines in areas where a low-permeability confining unit exists, and in the vicinity of withdrawal wells. This can result in aquifer yield overestimation. The current permitted withdrawal scheme in the study area based on the RAM2 solution overestimates sustainable yield by at least 30% as shown in the supplementary material (, 2025)[1]. Accordingly, using a distributed finite-element density-dependent flow model is needed to overcome these challenges along with an optimization algorithm to operationalize the sustainable yield policy in Hawai'i.

[1] Following Environmental Research: Water guidelines, this reference here and hereinafter is anonymized for double-anonymous peer review.

2.2 Simulation optimization problem

2.2.1 Method justification

In response to the need to improve the decision support tools related to sustainable yield management, we developed a simulation optimization tool to evaluate sustainable yield in Hawai‘i. Simulation optimization is a commonly used technique in groundwater management (Gorelick et al., 2015) that solves an objective function stating the management objectives aided by the simulation models and an optimization algorithm to find optimal management solutions. The management objectives, which were identified through stakeholder engagement, include minimizing saltwater intrusion and drawdown, and controlling spring discharge and submarine groundwater discharge. The Pearl Harbor aquifer system, which is an important aquifer system serving a population of about half a million people in Hawai‘i is used as a prototype.

The simulation optimization method was formulated based on a stakeholder driven modeling process using United States Geological Survey (USGS) developed models that are trusted by the stakeholder in the state. The calibrated and validated model of Pearl Harbor aquifer (Oki, 2005) was used to simulate density dependent groundwater flow. The simulation code is the finite element SUTRA model - Version 2.0 (Voss & Provost, 2002), which is a USGS model to simulate density dependent groundwater flow. We used a special version of SUTRA Version 2.0 (2D3D.1, Oki, 2005) that is specifically tailored for Hawai‘i to account for specific yield. The Hawai‘i Water Budget Model (HWBM, Engott et al., 2017) is used to generate recharge maps for different climate and land-use scenarios. Given the objective function, the simulation optimization procedure consists of the optimization algorithm CMA-ES (Hansen et al., 2003) that links the management objectives with the HWBM and the groundwater density dependent flow model. Since the simulation model is CPU intensive, the simulation optimization runs are carried out using parallel CMA-ES (Elshall et al., 2015) in high performance computing environment. We developed a simulation optimization code that has eight options to answer questions related to water allocation, spring discharge, climate change impact on recharge and sea-level rise, land-use scenarios, and economic analysis of withdrawal cost (, 2025).

This simulation optimization tool can be used to evaluate sustainable yield under different natural processes and societal preferences. Natural processes include the impact of climate change on top recharge, in-land lateral boundary recharge, and sea level rise. Two climate conditions are considered that are a baseline 1978–2015 historical climate (Giambelluca et al., 2014; Oki, 2005) and a dry future climate represented by the 2015-2065 RCP 8.5 mid-century projection (Timms et al., 2012), which is widely used in research and water-resources planning for Hawai‘i (Bremer et al., 2021; Brewington et al., 2019; Burnett

et al., 2020; Wada et al., 2017). Under RCP 8.5 midcentury rainfall, groundwater recharge is expected to decline by around 16% compared with baseline rainfall conditions, with the most pronounced reductions in high-elevation forested areas (Bremer et al., 2021). For future sea-level rise, we use a realistic framed scenario 0.5 m sea-level rise for the mid-century design period. This is an exploratory scenario describing not "what will happen" but "what could happen?"(Maier et al., 2016) to account for high uncertainty in sea-level rise projection. This is comparable to a mean value of 0.48 m sea-level rise in Honolulu area according to IPCC 6th Assessment Report Sea Level Projections in under RCP 8.5 in 2070 (NASA Sea Level Change Portal, 2025). Future inland lateral boundary recharge was estimated with linear regression given historical and future top recharge, boundary recharge, and pumping data (, 2025). Societal preferences include withdrawal management schemes that account for different ecological constraints on spring discharge and land-use scenarios. To quantify the tradeoff between spring discharge, submarine groundwater discharge and pumping, we ran the simulation optimization we consider five spring discharge scenarios that are non-constrained spring discharged, constrained spring discharge to current spring discharge flow rate, and constrained spring discharge to 40%, 60%, and 80% of the pre-development spring discharge flow rates. Land use scenarios include corridor and sprawl development with no-forest protection, targeted-forest protection, high-forest protection (Bremer et al., 2021). The simulation optimization procedure as presented in Fig. 1 can address different groundwater management questions related to sustainable groundwater withdrawal with respect to management decisions, climate change, landuse and landcover, spring discharge, and submarine groundwater discharge.

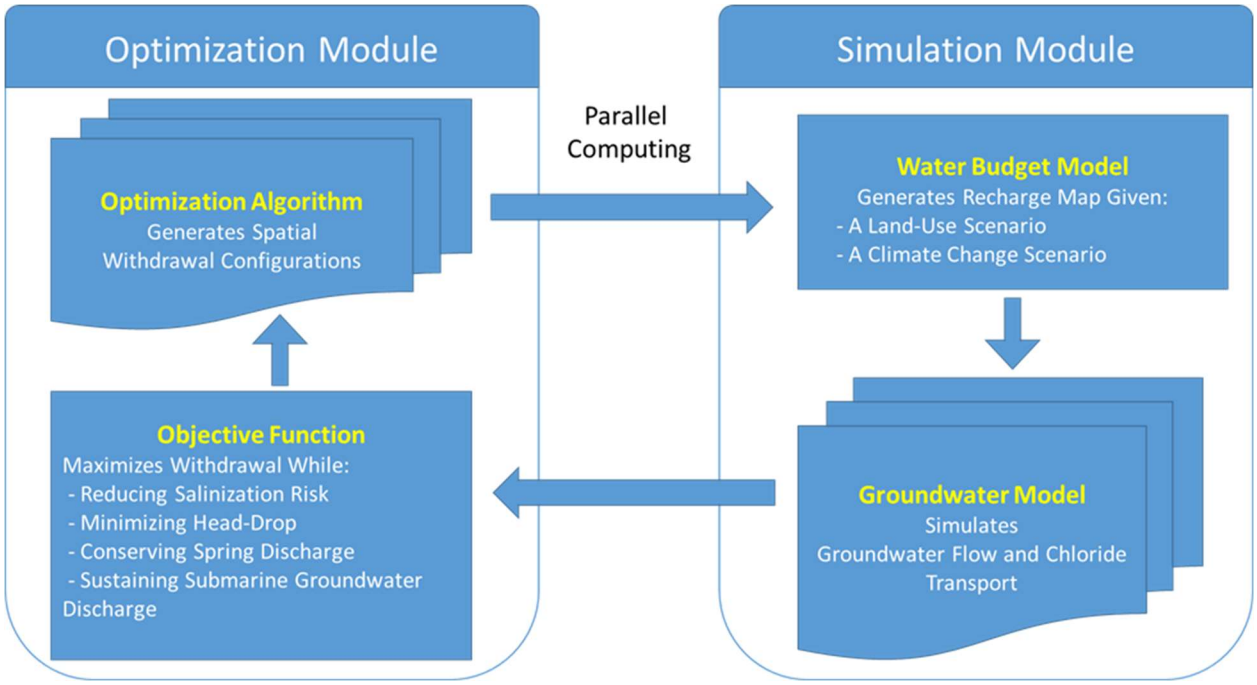


Fig. 1: Simulation optimization procedure to address different groundwater management questions

2.2. Problem formulation

Assuming linear combination of multiple objectives, the problem of finding an optimal solution can be stated as minimization function:

$$\min_{z\in\Omega_z} f(z) = \min_{z\in\Omega_z} \sum_{k=1}^{K} -f_k \quad (1)$$

Where f is the objective function that is generally formulated in a minimization convention; $\mathbf{z}(\mathbf{u},\mathbf{w})$ is a vector of decision variables that consists of a vector $\mathbf{u}$ of state variables and a vector $\mathbf{w}$ of decision variables; $\Omega_z = \Omega_u \cup \Omega_w$ is the feasible region of the decision variables $\mathbf{z}$ as represented by a set of constraint equations such that Ω_u represents the feasible region of the state variables $\mathbf{u}$, and Ω_w represents the feasible region of the decision variables $\mathbf{w}$; Given two objective functions $K = 2$, our state variables $\mathbf{u} \in \{u_1, u_2\}$ are the pumping wells chloride concentration and drawdown, and spring discharge, respectively. When $K > 1$ the problem is termed multi-objective optimization problem. Our decision variables $\mathbf{z}$ are withdrawal at pumping wells. Our constraints are the minimum or maximum thresholds, which the state variables $\mathbf{u}$ should not violate.

Our objective function is to find the maximum allowable withdrawal without violating these constraints. For any m decision variables of pumping wells, we used a withdrawal dependent objective function

$$f_1 = \sum_{i=1}^{m} r_i \times q_i \times p_i \qquad \begin{cases} p_i = -1 & \forall W_{i,obs} > W_{i,threshold} \\ p_i = 1 & \forall W_{i,obs} \le W_{i,threshold} \end{cases} \tag{2}$$

such that solution will be penalized if any of the chloride concentration or drawdown $W_{i,obs}$ in pumping well i exceeds chloride concentration or drawdown threshold $W_{i,threshold}$ for this pumping well; q_i is the withdrawal of pumping well *i*; and p_i is a penalty term. The weighting term r_i represents the relative importance of the pumping well *i* and we assume equal weighting for all decision variables. We observe the chloride concentration and drawdown $W_{i,obs}$ at the end of the simulation period. The spring objective function is a hard penalty constraint

$$f_2 = 4000 \times p \qquad \begin{cases} p = -1 & \forall S_{Obs} < S_{threshold} \\ p = 0 & \forall S_{Obs} \ge S_{threshold} \end{cases} \tag{3}$$

such that S_{obs} is the total spring discharge observed at the end of simulation period; $S_{threshold}$ is the spring discharge threshold for all springs combined; p is a penalty term for total spring discharge. The parameterization of this simulation optimization problem such selection of decision variables, design period, and threshold values can be conducted through technical experts or through participatory modeling (section 4).

3. Collaborative modeling for simulation optimization problem formulation

3.1 Overview

Collaborative modeling tools were developed through several activities, including a participatory approach with multiple government agencies to design land use scenarios (Bremer et al., 2021), and social learning activities with community members to better understand relational values and prioritize important ecosystem services that are related to spring discharge (Burnett et al., 2020). To address the complexities of groundwater sustainability, this study introduces a collaborative modeling framework that integrates and extends the guiding principles of participator modeling in water resources (Langsdale et al., 2013) with the standardized reporting method of Gray et al. (2018) as illustrated in Fig. 2. The reporting approach of Gray et al. (2018) has four components that are purpose (why), process (how), partnership (who), and products (what). The first three of which are presented in this section. The fourth component is presented in the results and discussion section (Section 4.2).

Collaborative modeling	
Reporting	**Guiding Principle**
Purpose (Why)	1. Collaborative modeling is appropriate for complex, conflict-laden, decision-making processes where stakeholders are willing to work together 2. The model addresses questions that are important to decision makers and stakeholders
Process (How)	3. The model supports the decision process by easily accommodating new information and quickly simulating alternatives 4. Collaborative modeling builds trust and respect among parties
Partnership (Who)	5. All stakeholder representatives participate early and often to ensure that all their relevant interests are included 6. Collaborative modeling requires both modeling and facilitation skills
Product (What)	7. Both the model and the process remain accessible and transparent to all participants. 8. Parties share interests and clarify the facts before negotiating alternatives.

Fig. 2 Collaborative modeling methodological framework to develop simulation optimization tool for decision support in groundwater sustainability based on the guiding principles of Langsdale et al. (2013) and standardized reporting approach of Gray et al. (2018).

3.2 Purpose

Following the reporting approach in Fig. 2, we described the participation purpose and the model purpose alongside *Principle 1*. We included stakeholders in the modeling process for several purposes. First, participation is an integral policy component in Hawai‘i in that the Commission on Water Resource Management is obliged to cooperate with agents for “the purpose of utilizing and conserving the waters of the State, and assist these organizations and agencies in coordinating the use of their facilities and participate in the exchange of ideas, knowledge, and data with these organizations and agencies” (§174C-5, HRS Chapter 174C, 1987). Second, this participation process is imperative to attain an adequate qualitative overview of the problem, and a shared vision of the problem with stakeholders (Martinez-Santos et al., 2008; Molina et al., 2011). Third, collaborative modeling as a form of participation can lead to more effective solutions being identified and adopted (Watson, 2005). This is mainly because participation involves tapping into institutional knowledge, exchange of experiences, deeper understanding, consensus building, and raising commitment toward resource management (Carr et al., 2012; Castilla-Rho, 2017; Mays, 2013). Stakeholder input in this case is broadly useful, ranging from understanding historical factors and local knowledge, to devising realistic modeling assumptions, scenarios, and management alternatives. In addition, collaborative modeling makes the developed simulation optimization framework applicable to the stakeholders’ decision-making

context, and ensures that the knowledge and needs of the stakeholders are incorporated in the simulation optimization framework (Gray et al., 2018). Additionally, collaborative modeling allows stakeholders to vet, correct and improve modeling assumptions. Finally, collaborative modeling is an effective uncertainty analysis method. As shown below, this collaborative modeling approach is helpful for dealing with uncertainty related to the complexity of the socioeconomic system that is difficult to address otherwise.

The adopted simulation optimization framework has several purposes and multiple problems to address. According to *Principle 2* as shown in Fig. 2, collaborative modeling prioritizes designing the model based on its intended purpose, rather than developing a technically driven model first and finding users afterward (Langsdale et al., 2013). As each model has specific purposes, understanding the model's purposes is important to decide on the model scope and level of analysis. With respect to improving hydrologic and environmental decision-making in the context of groundwater sustainability in Hawai‘i, the main purpose of this simulation optimization framework is the proof of concept of the value of numerical modeling within a simulation optimization framework to evaluate groundwater sustainability. The use of an analytical model such as RAM2 versus a numerical model such as SUTRA is a highly debated topic as reviewed by (Elshall, et al., 2020). Analytical models can be inadequate for evaluating groundwater sustainability (Henriksen et al., 2008; Kalf & Woolley, 2005; Mulligan et al., 2014) because they lack the ability to simulate various leakage components, their treatment of the interaction between inflow and outflow components is often not rigorous, and they are typically not designed to consider spatiotemporal relationships. While analytical models are subject to criticism, the choice of analytical model versus numerical model is case-specific and depends on data availability, sustainability factors of interest, and aquifer type (Elshall, et al., 2020). Yet apart from the advantages and disadvantages of analytical models, there is a need for a tool to evaluate groundwater sustainability in Hawai‘i that can flexibly account for hydrological systems, groundwater-dependent ecosystems and human activities under different natural and anthropogenic stresses. This is particularly needed as serious challenges of groundwater sustainability in coastal aquifers are arising due to climate change and socioeconomic development (Michael et al., 2017).

The situation in Hawai‘i and similar islands is more critical due to their special hydrological features and sea level rise combined with anthropogenic impacts causing water stress (Werner et al., 2017). Evaluating groundwater sustainability with respect to these challenges requires integrating hydrologic models with different constraints such as socioeconomic demographics, human activities, societal preferences, groundwater dependent ecosystems, climate change impacts and mitigation, and regulations, among other aquifer performance and governance factors (Alegria et al., 2024; Chen et al., 2025; Cuthbert et al.,

2023; A. S. Elshall, Arik, et al., 2020; Gleeson et al., 2020; Karandish et al., 2025; Perrone et al., 2023; Pierce et al., 2013; Rohde et al., 2024; Sadeghi-Jahani et al., 2024; Wang et al., 2024). Illustrating the essential factors to consider when evaluating groundwater sustainability, as shown in Fig. 3, can be useful in collaborative modeling to understand the scope of the evaluation. Several of these factors are considered in this study. With respect to recharge rates and storage conditions we consider climate change, land use and land cover change impacts, drawdown, natural and induced recharge, and storage. With respect to water quality, we consider saltwater intrusion. With respect to capture, discharge rates, and environmental flows we consider beneficial usages for springs and submarine groundwater discharge. With respect to facilities and technologies we consider pumping, nature-based infrastructure, and desalination for groundwater substitution. With respect to legal and institutional constraints we consider water rights, restrictions on production well locations and depths, restrictions on specific activities, and sustainable yield policy. With respect to societal values and preferences we consider mainly instrumental, intrinsic, relational and aesthetic values. With respect to economic feasibility, we consider and discuss groundwater substitution and pumping costs, although these factors are not quantitatively incorporated into the model. Combining these different aquifer performance and governance factors requires a modular tool. We demonstrate how this modeling framework can better address major concerns such as incorporating aquifer governance parameters related to groundwater usage and accessibility, spring discharge, sea level rise, and future recharge scenarios into groundwater sustainability evaluation in Hawai‘i.



Fig. 3. Factors to consider in groundwater sustainability evaluation modified from (Elshall, et al., 2020).

3.3 Process

Process reporting includes reporting the level of participation, interaction between the participants and the model, and the relationship between the collaborative modeling and the decision-making process (Gray et al., 2018; Langsdale et al., 2013). We used collaborative modeling to formulate the simulation optimization problem. Collaborative modeling refers to a high degree of participation and cooperation between the modeling team and stakeholder. Given the ladder of participation and types of cooperation chart proposed by Basco-Carrera et al. (2017), our approach is a discussion and co-design type of participation that is more than just consultation, but less than co-decision making, and a collaboration type of cooperation that is more than just unilateral action or coordination, but less than joint-action as shown in Fig. 4. Given this level of participation, the simulation optimization framework is constructed based on the stakeholder knowledge, which increases the degree of model transparency and the stakeholder understanding of model assumptions.

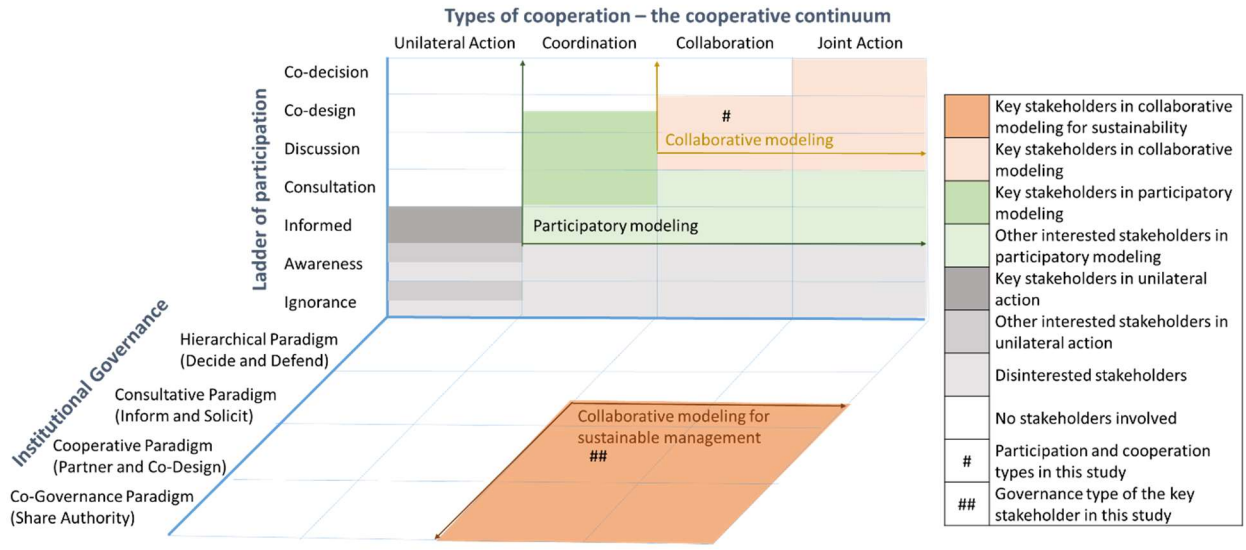


Fig. 4. Framework for stakeholder engagement integrating institutional governance of key stakeholders with the type of participation and cooperation, extended and modified from the ladder of participation and types of cooperation chart proposed by Basco-Carrera et al. (2017).

The participants interact with a purely technical version of the model, which acts as a boundary object to facilitate the co-design of the modeling framework with stakeholders. The *Principle 3* in Fig. 2 focuses on integration and resolution of many issues rather than the depth and precision, and thus keeping the model relatively simple and at a high level is recommended at this stage, and then detailed and precise analysis can be conducted after prioritizing options (Langsdale et al., 2013).The modeling team developed a simulation optimization procedure from a purely hydrologic perspective regardless of the social context. Thus, assumptions about decision variables, initial conditions, observation points, among other settings were set based on the research team's general modeling experience, rather than hands-on experience about aquifer management. For example, our decision variables are either selected pumping wells or pumping clusters. Optimizing withdrawal of the 92 pumping wells in the study area is a high dimensional problem. This is computationally unfeasible since the iteration size and the number of iterations to reach convergence increase with the number of decision variables . Also, having a high dimensional problem is not technically recommended since a large number of decision variables can cause solutions non-uniqueness, resulting in multimodality. A common approach to reducing the problem of dimensionality is to group pumping wells into clusters such that the number of decision variables is the number of pumping clusters instead of the number of pumping wells (Pholkern et al., 2019). We grouped the 92 pumping wells into 20 clusters based on spatial proximity using the k-mean clustering method as shown in supplementary material (, 2025). Although this is a reasonable

assumption, it does not tap into the local knowledge of aquifer managers. Similarly, we made reasonable assumptions about other settings later will be discussed Table 1 and obtained simulation optimization solutions that serve as a starting point for the co-design process.

We presented the developed simulation optimization method and solution to promote collaborative learning in line with Principle 4 in Fig. 2. The meeting agenda was designed to first introduce the project and modeling assumptions and then ask participants to contribute to the model development through providing data, improving model assumptions, validating outputs, prioritizing important scenarios, and providing insights from real-life experiences about the aquifer. Langsdale et al. (2013) note the importance of positive relationship building, beyond simply bringing people together. In particular, when emotional participants tend to be more open to discussion and accepting new ideas. Additionally, involving trusted and respected community members in the modeling team can greatly aid positive relationship building. In this case study, these factors helped to develop a shared or co-learning approach, which is a form of collaborative learning, where information flows from research team to stakeholder and vice versa (Basco-Carrera et al., 2017). As such, the research team presented the simulation optimization method, problem formulation, solution and capabilities to the stakeholder for feedback. The resulting discussions, which tapped into the stakeholder's collective knowledge about the aquifer, supported the co-design of the simulation optimization problem. The stakeholder's collective knowledge includes (1) well-rounded knowledge about aquifer stakeholder regulations including relationships with other stakeholders, (2) hands-on experience about aquifer management including experience about the most relevant hydrogeological features of the aquifer and social dynamics of the aquifer management, and (3) an in-depth understanding of aquifer users' perceptions, concerns and preferences. Such collaborative learning assists in the acquisition of collective skills and increases individual knowledge within the social context (Basco-Carrera et al., 2017). In addition, this collaborative learning and co-design directly relates the modeling framework to the decision context.

3.4 Partnership

Partnership reporting involves reporting the participant selection process, and the steps participants are involved with. Identifying stakeholders and defining avenues for participation is largely contextual (Carr et al., 2012; Kusters et al., 2017). The stakeholder selection process in this case study is straightforward as there is no need to extend the collaborative modeling beyond key stakeholders. As connections are made first, defined by the Hawai‘i (HRS Chapter 174C, 1987), the main stakeholder with respect to sustainable yield evaluation in the state is the Commission on Water Resource Management (CWRM), which is the state water regulator. According to *Principle 5* (Fig. 2) all stakeholder representatives should engage early and frequently to ensure that their relevant interests are adequately represented. CWRM acts as a representative for community members and entities who have interest and concern about groundwater as it includes members most directly involved with the model building team, communicating with key stakeholders, who

act as a trusted link to interested parties. Thus, the CRWM inputs are intended represent the interests of the wider community, and alternatives are evaluated, while attempting to balance the many diverse interests, to identify the most beneficial decision. A second important stakeholder is the local water manager in the study area, Board of Water Supply (BWS), a semi-autonomous agency that manages O‘ahu's municipal water resources and distribution system.

The nature of the partnership, as defined by institutional governance, is a key dimension when considering sustainable groundwater management or sustainability in general. This dimension, as shown in Fig. 4, captures the fundamental governance character of a stakeholder, ranging from the hierarchical paradigm of a technocratic institution with a strong engineering and regulatory mandate to the deeply stewardship-oriented and consensus-driven approach that, for example, can be found within Indigenous communities. In this study, CWRM exemplifies cooperative governance (Fig. 4), as CWRM is mandated by the State Water Code to "participate in the exchange of ideas, knowledge, and data" with other agencies and organizations. This pre-existing collaborative paradigm with public is a critical enabling factor for creating a receptive environment for the co-design of sustainable groundwater management. The distinction is crucial, as high-level engagement like co-decision making may still fail to produce sustainable outcomes if the institutional partner has a hierarchical paradigm instead of genuinely cooperative governance style as illustrated in Fig. 4.

Participants are involved through multiple steps, starting with informal meetings and communication to inform stakeholders about the project, assessing the avenues and level of participation that they are interested in, and obtaining updated pumping data to start designing the technocratic version of the simulation optimization framework before the collaborative modeling process. The collaborative modeling process involved a full-day workshop to co-design the simulation optimization framework. According to Principle 6 (Fig. 2) of Langsdale et al. (2013), collaborative modeling requires the integration of both technical modeling expertise and facilitation skills. The role of the lead modeler was to lead technical discussions, distill expectations, synthesize the inputs of the stakeholders, and build a co-designed model with the stakeholders. The role of the facilitator was to contact stakeholders to plan for the workshop and facilitate discussion during the workshop. This workshop was followed by a meeting to present the updated simulation optimization results based on the co-design process. These two events were attended by the modeling team and CWRM staff. At the project completion a third meeting occurred, which was attended by CWRM, BWS, and the modeling team. The objective of the meeting was to obtain stakeholder feedback on the modeling output and discuss future steps.

4. Results and discussion

4.1 Product

The formulation of the simulation-optimization problem was refined through an iterative, collaborative process, with the key outcomes of the co-design process summarized in Table 1. This engagement directly shaped the structure and assumptions of the simulation optimization to reflect practical management realities.

The primary outcome of the collaborative modeling was the redefinition of the decision variables of the simulation optimization. Stakeholders advised against the initial, purely technical approach of k-means clustering. Instead, they recommended a more targeted strategy that focused the optimization on individual wells with pumping rates exceeding one million gallons per day (mgd). This simplified the optimization problem while retaining the most managerially relevant wells. To align with existing governance frameworks, wells were grouped by interest groups and administrative aquifer units as shown in Fig 5 versus the original clustering scheme as shown in the supplementary materials (, 2025). Pumping wells were categorized according to three main interest groups that are (1) Board of Water Supply (BWS) wells and shafts, (2) federal wells, and (3) other wells such as state government wells, irrigation wells, home associations wells, wells with missing information, and other private wells. Pumping wells were also categorized by aquifer administrative units (i.e., Ewa-Kunia, Waipahu-Waiawa, Waimalu and Moanalua) according to the Department of Land and Natural Resources (DLNR). Thus, to be consistent with the current management practice, we report on the sustainable yield for each administrative unit. The modeling team and the stakeholders co-designed the decision variables selection based on interest groups and current withdrawals. The current withdrawal of BWS wells and shafts, federal wells and other wells are 90.25, 19.07 and 7.63 mgd, respectively, with a total of 117 mgd. The current permitted withdrawal of BWS wells and shafts, federal wells and other wells are 124.97, 22.67 and 33.46 mgd, respectively, with a total of 181 mdg. Wells were categorized by current withdrawal >0.01 and ≤0.1 mgd (48 wells), >0.1 and ≤1 mgd (44 wells), and >1 mgd (27 wells). Stakeholders suggested to start by optimizing withdrawal for pumping wells with current withdrawal above 1 mgd. This co-design did not eliminate the clustering uncertainty but also encoded the stakeholder knowledge about the system in the simulation optimization.

A second outcome was to co-design the objective function of the simulation optimization by encoding the management objectives that are shown in Figure 1. With respect to reducing salinization risk, a critical outcome of this co-design process was the implementation of a dual-salinity threshold with a relaxed standard of 1,000 mg/l chloride for caprock wells used primarily for irrigation and a stricter 250 mg/l standard for basalt aquifer wells supplying

potable water. Also, they confirmed that observations at decision-variable wells with pumped-well mixing for chloride and evaluation at end-of-simulation are appropriate assumptions. With respect to minimizing drawdown, the stakeholders recommended that the head-drop limit to less than or equal 1 m from pre-development as good management practice. With respect to conserving spring discharge and maintaining submarine groundwater discharge (SGD), in response to public pressure to SGD, the stakeholders advised that the simulation optimization can treat spring discharge as a hard constraint while tracking SGD for transparent trade-off reporting.

The co-design process was also crucial for establishing and validating key technical assumptions and constraints. Stakeholders agreed that using period-average withdrawal for initial conditions and observing outcomes at the end of the 50-year simulation from 2015 to 2065 was a reasonable approach. For the initial condition and decision-variable design, stakeholders endorsed representing “current withdrawal” for 2001–2015 as a period average, with the explicit caution that missing entries must not be imputed as zeros; using un-averaged current data was also considered acceptable. They recommended constant (time-invariant) withdrawals over the design horizon for interpretability and advised omitting per-well min/max bounds in initial runs, with the option to add bounds later in consultation with BWS.

Collectively, these settings operationalize stakeholder priorities in the simulation–optimization (50-year horizon) keeping the final model outputs aligned with management practice and communication needs. The co-design process surfaced and resolved numerous modeling choices (e.g., averaging windows, observables, constraint placement, formation-specific salinity thresholds, and the choice to forgo clustering) that would otherwise remain implicit “under the carpet.” Making these assumptions explicit, negotiated, and documented not only improves scientific transparency and reproducibility but also increases the salience and legitimacy of the results for decision-makers. In other words, this co-design process ensured the final model was not only technically robust but also salient to the specific needs and context of groundwater management in the region. For the result of the paper, we will provide selected model outcomes to discuss how the model can be further used for stakeholders’ engagement.

Table 1: Summary of the co-design process in the first workshop. The underlined text represents the final, consensus-based assumptions

Topic	Questions and feedback
Initial condition	Q: Current withdrawal for prediction period from 2001 to 2015: (i) monthly average, (ii) annual average, (iii) period average or (iv) other? Period averaging is a reasonable assumption, and missing data should not be included as zeros in the averaging. Using that current data without any averaging is also a reasonable choice
Decision variables	Q: Pumping wells: (i) selected pumping wells, (ii) pumping clusters or (iii) all pumping well Clustering can be avoided by selecting relevant pumping wells such as the Board of Water Supply (BWS) wells and the Navy wells. Private wells, especially those with low withdrawal rates, do not need to be included in the decision variables
	Q: Time-dependent withdrawal: (i) constant withdrawal or (ii) time variant withdrawal Constant withdrawal is a reasonable assumption as time-variant withdrawal is not needed and it is difficult to present and communicate
	Q: Withdrawal rates: (i) minimum and maximum withdrawal per pumping well if any or (ii) a uniform distribution from zero to a large number For initial test runs defining minimum and maximum withdrawal per pumping well is not needed, yet that can be later incorporated based on feedback from BWS
Clustering	Q: Withdrawal clusters: (i) K-mean clustering, (ii) aquifer units, or (iii) other Clustering can be avoided, yet pumping wells can be categorized by (i) DLNR aquifer units, (ii) Interest groups (e.g., BWS wells, federal wells, irrigation wells, private wells, etc.) and (iii) aquifer property (e.g., basalt aquifer, Caprock, valley-fill barrier, etc.). Salinity threshold at the Caprock wells should be 1000 mg/l chloride versus 250 mg/l for the basalt aquifer
	Q: Withdrawal ratios for pumping wells in each cluster: (i) current, (ii) permitted, (iii) evenly distributed or (iv) other Using the current ratio is a reasonable assumption, but this is not needed if we will avoid clustering
	Q: Withdrawal rates for the fixed pumping wells: (i) permitted, (ii) current or (iii) zero Using the current rates is a reasonable assumption
	Q: Current withdrawal: (i) monthly average, (ii) annual average, (iii) average from 2001 to 2015 or (iv) other Using an average withdrawal from 2001 to 2015 is a reasonable assumption
Observation	Observation locations (i) decision variable pumping wells, (ii) decision variable pumping wells plus additional observation locations, (iii) other? The current selection is reasonable and additional observation locations can be included depending on feedback from BWS
	Observed salinity: (i) per node, (ii) water mixing per pumping well or (ii) water mixing per cluster? The current selection of water mixing per pumping well is reasonable
	Observation time: (i) Observation at the end of simulation, (ii) observation at time shift (maximum), (iii) observation at time shift (average)? Observation at the end of the simulation is a reasonable assumption as it is not stringent and reflects the desired situation at the end of the design period in 2070
Additional feedback	Other points? There is public pressure on CWRM and BWS to report on the impact of withdrawal on submarine groundwater discharge

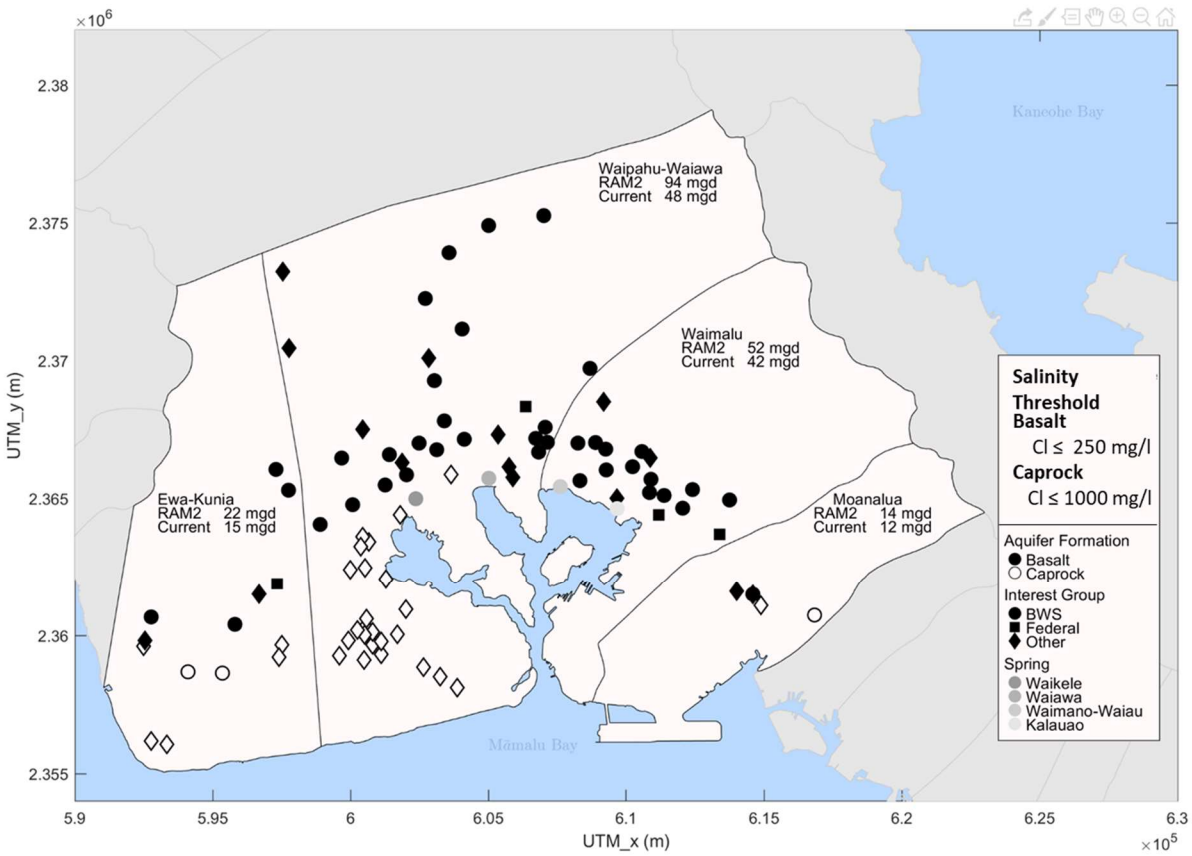


Fig. 5. Updated pumping well map based on stakeholder engagement, in which pumping wells are categorized by aquifer property (basalt and caprock), aquifer unit (Ewa-Kunia, Waipahu-Waiawa, Waimalu and Moanalua) and interest group (BWS, Federal and Private).

4.2 Mid-century projection without optimization

Under the dry mid-century climate, recharge declines (both top and boundary) intensify saltwater intrusion, reducing the volume of fresh groundwater that can be sustainably pumped even if current pumping scheme is constant. Relative to current conditions, your projections show a 12% loss of fresh withdrawal capacity under present climate variability due to salinization as shown in Fig. 6.

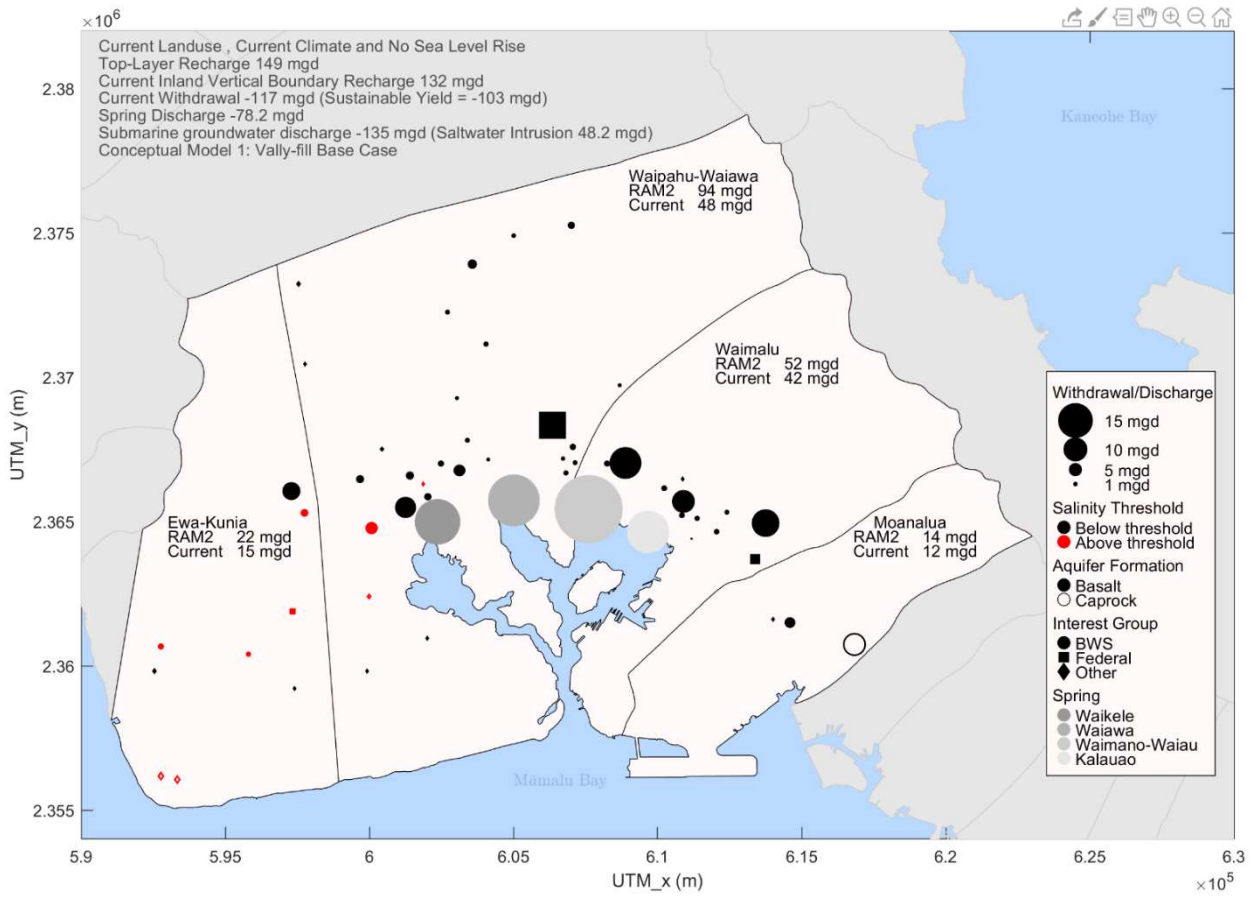


Fig. 6. Future projection of current withdrawal rates under current climate of both recharge and sea-level rise with current land-use management.

When future drying is imposed, the freshwater loss increases to 26%, and when sea-level rise is added the sustainable withdrawal collapses from 117 to 63 mgd (−46%). If land-use change follows a sprawling, no-forest-protection trajectory, the decline reaches 48%, demonstrating that climate-driven recharge reductions and sea-level rise interact with land management to shrink the potable portion of the resource and push more wells above salinity thresholds (red markers) as shown in Fig. 7, raising clear implications for water-supply reliability. The simulation optimization can provide mitigation solutions for the impact of climate change through optimizing pumping. Moreover, simulation optimization provides a framework for evaluating other migration options including nature-based infrastructure such as corridor development and watershed restoration through target and high-level forest protection. In the remaining of the paper, we will show how collaborative modeling can improve the credibility, legitimacy, and saliency of this modeling process.

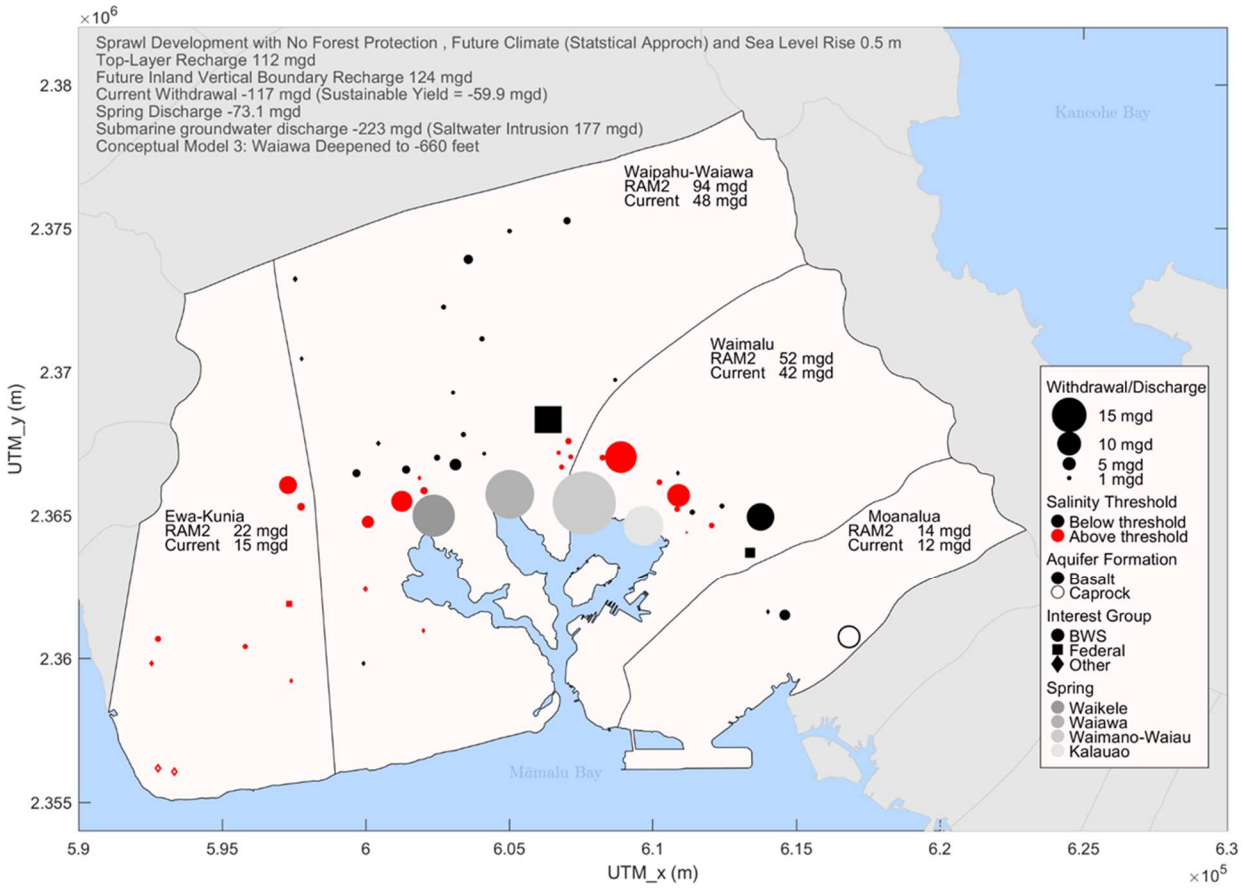


Fig. 7. Future projection of current withdrawal rates under future climate projections of both recharge and sea-level rise with suboptimal land-use management.

4.3 Withdrawal management scenarios

The optimization code can accommodate various withdrawal schemes to account for different water withdrawal scenarios that the stakeholder may be interested in evaluating. The stakeholder mentioned that BWS wells should be prioritized as decision variables, while federal wells are generally managed separately. Stakeholders advised that decision variables should wells with average withdrawal from 2001 to 2015 larger than 1 mgd. We evaluate two withdrawal management scenarios that were suggested by stakeholders. The withdrawal management scenario of including federal wells results in 27 decision variables (22 BWS wells, 3 federal wells and 2 other wells) and not including federal wells results in 24 decision variables (22 BWS wells and 2 other wells). Other wells are fixed at their current withdrawal.

Simulation optimization can mitigate saltwater intrusion due to non-optimal pumping rates. When including federal wells, the projected fresh groundwater availability under an

optimized pumping strategy that constrains spring discharge to current levels is presented in Fig. 8. The sustainable yield estimate of 126 mgd represents an 8% increase compared to the projection based on current conditions (Fig. 6). Note Fig. 8 shows optimal maximum withdrawal of 127 mgd and sustainable yield estimate of 126 mdg. We make this nuance distinction to clearly illustrate the impact of pumping optimization on other wells that are fixed and not part of the optimization process. Wells with withdrawal less than 0.1 mgd are not shown. Submarine groundwater discharge (SGD) and saltwater intrusion refer to outflow and inflow at the coastal boundary, respectively. RAM2 estimated withdrawal (RAM2), current withdrawal (current), and simulation optimization maximum withdrawal (optimum) are shown for each administrative unit. Fig .8 shows for all the aquifer administrative units, the estimated maximum allowable withdrawal is greater than or equal to current withdrawal. Results indicate that the optimal pumping schedule can effectively reduce saltwater intrusion, which was the driver behind the 12% decline in the non-optimized fresh groundwater projection (Fig. 6).

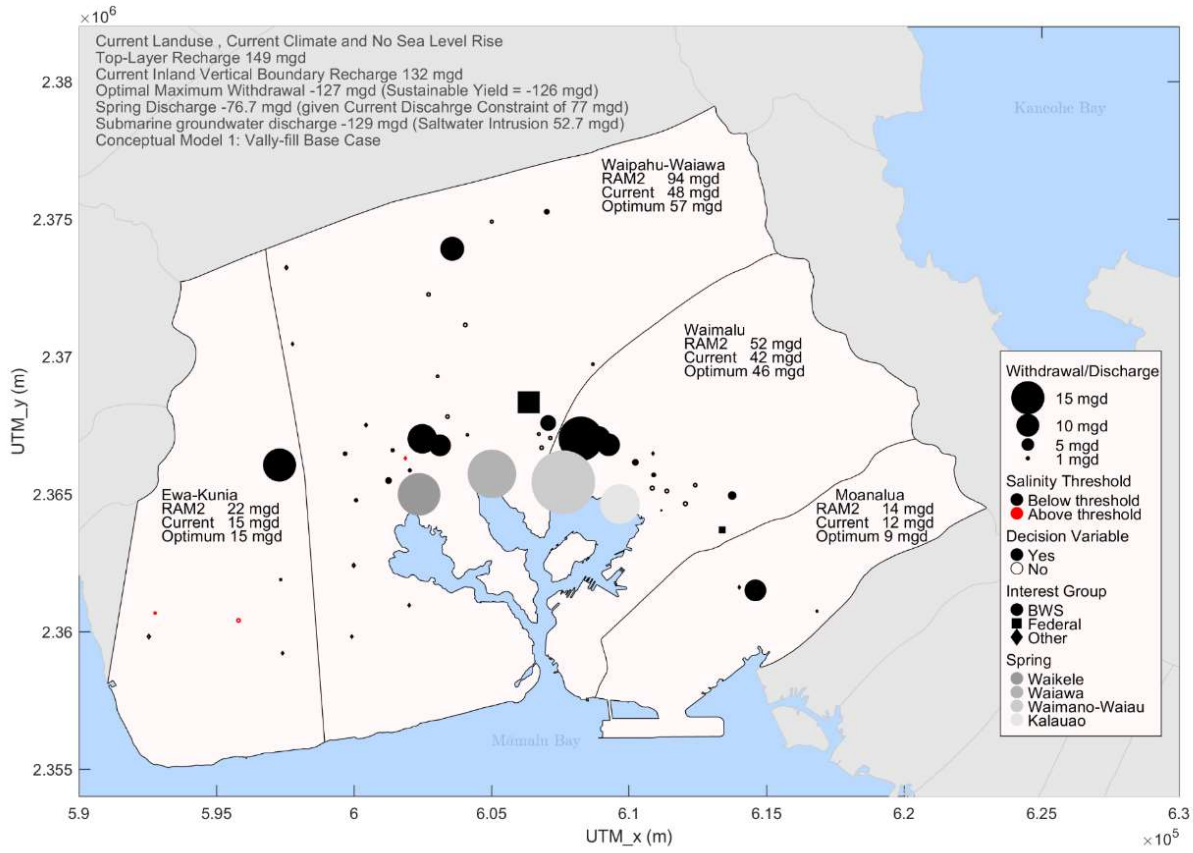


Fig. 8. Projected groundwater availability under an optimized pumping strategy that constrains spring discharge to current levels under current climate of both recharge and sea-level rise with current land-use management.

For withdrawal scenario that does not include federal wells as decision variables, the maximum allowable withdrawal is reduced to 124 mgd as shown in the supplementary materials (, 2025). The impact of increasing the number of decision variables will increase the degrees of freedom and thus may result in higher maximum allowable withdrawal. The inclusion of the federal wells provided added flexibility of 2% in optimized groundwater withdrawal. This small difference indicates that the inclusion or exclusion of federal wells has little effect on sustainable groundwater availability.

This finding adds significant practical value and legitimacy to the optimization results and would not have been possible without the stakeholder-driven approach. In addition, collaborative modeling process made the distinction between groundwater availability (i.e., the physical capacity of the aquifer) and groundwater accessibility (i.e., the institutional and administrative ability to use it) explicit and testable. A purely technical optimization (e.g., all-wells or clustering-based optimization) might have overlooked this institutional nuance, which could result in overstating gains, obscuring administrative trade-offs, and yielding solutions that are infeasible to implement. Thus, collaborative modeling adds value by providing transparent rationales for why including or excluding federal wells does not change the practical availability of groundwater to the municipal system to inform public policy.

4.4. Pumping optimization for climate impact mitigation

Simulation optimization can mitigate saltwater intrusion due to non-optimal pumping rates and climate. Projected fresh groundwater. As suggested by stakeholders, we evaluate the separate and combined impacts of different climate change components, which are in-land vertical recharge, top recharge and sea-level rise, to understand to understand to what extend pumping optimization can mitigate these different future conditions. The projected freshwater availability under this separate and combined impact with optimized groundwater pumping is presented in Table 1. These results highlight the effectiveness of pumping optimization in reducing the impacts of decreased recharge and increased saltwater intrusion. For example, while the combined effects of future recharge and sea-level rise reduce the optimized sustainable yield to 99.4 mgs, this is a significant improvement over the 63 mgd sustainable withdrawal projected without optimization. Table

2 also shows that sea-level rise is a primary driver of saltwater intrusion under future conditions even with optimized pumping. In addition, stakeholder-defined spring constraints determine what portion of that physical availability is accessible for policy. When the spring discharge is constrained to the current rate, optimized sustainable yield declines from 126 mgd (current climate) to ~101 mgd under future recharge scenarios (20% drop), underscoring that induced recharge loss from spring discharge, not sea-level rise alone, drives the largest reductions. In other words, ecological flow requirements appropriately translate some "availability" into non-pumpable, protected outflows.

Table 2. Projected groundwater availability given current landuse and landcover scenario with current top recharge (149 mgd), future top recharge (132 mgd), current in-land recharge (132 mgd), future in-land recharge (124 mgd), current sea-level (0 m), future sea level (+0.5 m) without constraining spring discharge and with constraining spring discharge to current rate (77 mgd).

Top Recharge	In-land Recharge	Sea-level	Spring Discharge Constraint	Optimal Maximum Withdrawal (mgd)	Sustainable Yield (mgd)	Spring Discharge (mgd)	Submarine groundwater discharge (mgd)	Saltwater Intrusion (mgd)
Current	Current	Current	No	-155	-152	-61.8	-119	56.6
Current	Current	Future	No	-145	-141	-75.9	-234	176
Current	Current	Current	Current	-127	-126	-76.6	-129	52.7
Future	Current	Current	No	-129	-127	-62.4	-115	55
Future	Current	Future	No	-115	-109	-79.6	-234	175
Future	Current	Current	Current	-104	-101	-76.6	-126	52
Future	Future	Current	No	-120	-115	-58.6	-113	55.4
Future	Future	Future	No	-102	-99.4	-77.6	-231	175
Future	Current	Current	Current	-103	-101	-76.8	-126	51.3

The findings illustrate the value of collaborative modeling. Separate and combined evaluation of the climate stressors into separate and combined impacts was suggested by stakeholders. A purely technical exercise might have focused only on the final, combined-impact scenario. By testing the scenarios stakeholders prioritized, the modeling process provided a transparent and salient breakdown of risks. This allows managers to see the relative impact of each climate component such as reduced recharge versus sea-level rise on sustainable yield, thereby building trust and providing a much clearer, more legitimate

basis for policy discussions and adaptation planning. This nuanced understanding would not have been possible without co-design.

4.5. Nature-based solutions for climate impact mitigation

Land-use and watershed management practices play a crucial role in shaping groundwater sustainable yield, particularly under projected climate change stressors. Land cover change can influence recharge and sustainable yield, though is likely a less significant driver of groundwater recharge than climate change in the study area (Bremer et al., 2021). As presented in Fig. 9, high forest protection can help mitigate the impacts of reduced top-layer recharge from climate change and prevent land-use changes from compounding these negative effects. While the difference between corridor development (9.b-d) and sprawl development (9.e-g) scenarios is modest in terms of pumping volume, sprawl development introduces more negative socio-environmental consequences, including habitat fragmentation, water and air pollution, increased infrastructure costs, inequality, and social homogeneity. Scenario b in Fig. 9 for corridor development with high forest protection shows a notable increase of 32 mgd in total pumping capacity compared to scenario a (current conditions), indicating the hydrological value of nature-based solutions and pumping optimization. With optimization alone, 21 mgd more than current withdrawal can be sustainably pumped, but incorporating forest protection allows for an additional 11 mgd, highlighting the synergistic benefits of combining pumping strategies with watershed restoration. These results affirm that nature-based solutions like forest protection are a cost-effective and scientifically supported strategy to enhance climate resilience in groundwater management.

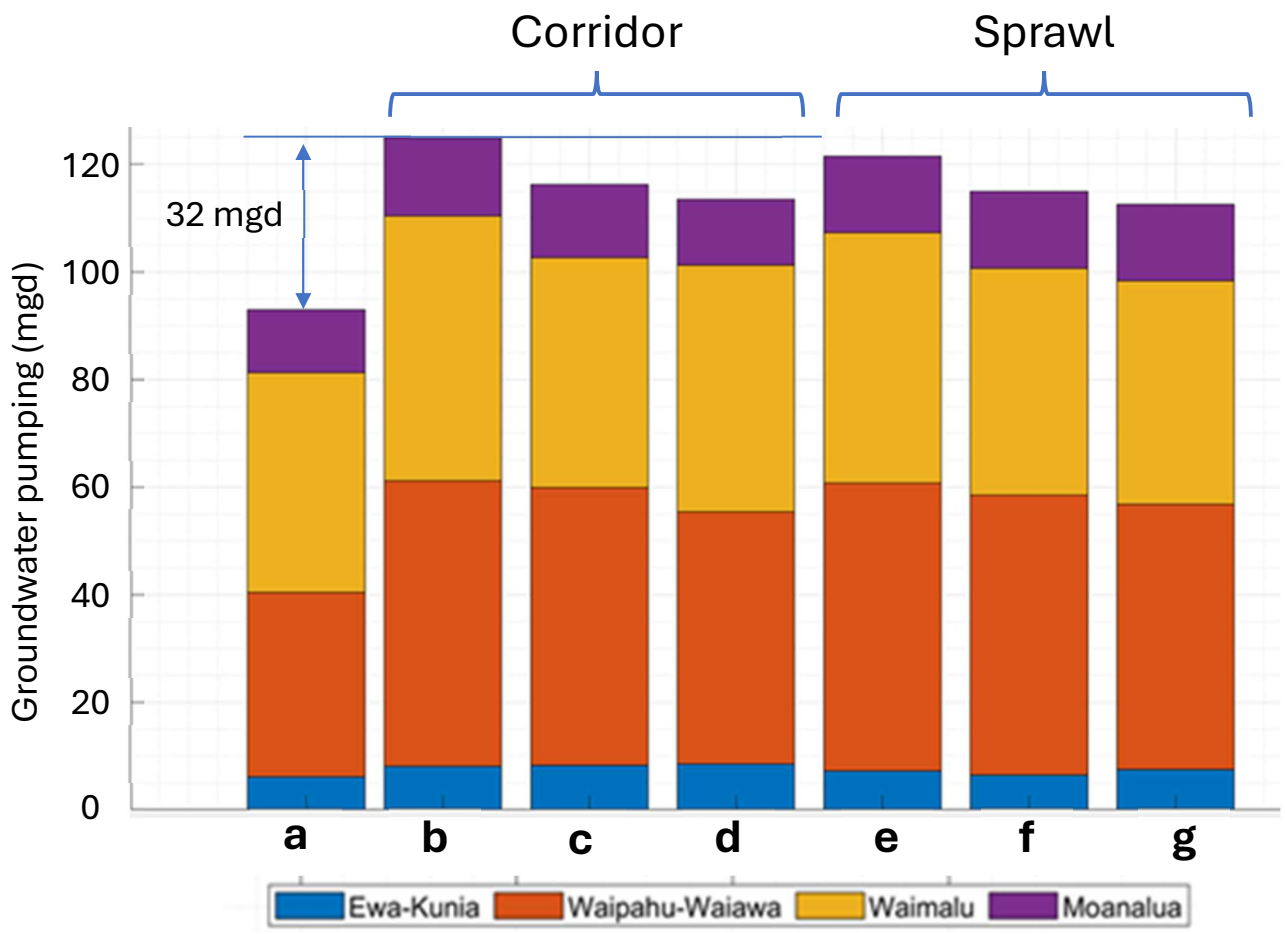


Fig. 9. Projected groundwater pumping under six landuse and landcover scenarios, combining two development patterns and three forest protection levels: (b–d) Corridor development with high, targeted, and no forest protection; (e–g) Sprawl development with high, targeted, and no forest protection. (a) represents the projection of current condition.

We further investigate corridor development with high level forest protection under current spring discharge constraint of 77 mgd with three repeat simulation optimization runs (Table 3). The results from these runs are highly consistent, with Runs 2 and 3 converging to an identical sustainable yield of 96.8 mgd and Run 1 yielding a very close 98.9 mgd. This demonstrates the robustness of the optimization algorithm and the reliability of the solution.

Table 3. Projected groundwater availability given high level forest protection and future top recharge (121 mgd) with constraining spring discharge to current rate (77 mgd).

Simulation Optimization Run	Optimal Maximum Withdrawal (mgd)	Sustainable Yield (mgd)	Spring Discharge (mgd)	Submarine groundwater discharge (mgd)	Saltwater Intrusion (mgd)
Run 1	-101	-98.9	-76.8	-125	50.9
Run 2	-99.1	-96.8	-76.7	-127	50.3
Run 3	-99.1	-96.8	-76.7	-127	50.3

These findings show the value added by the collaborative modeling framework. Developing and evaluating scenarios with stakeholders is effective method in simulation-optimization studies to explore trade-offs and evaluate management alternatives (Raei et al., 2017; Fowler et al., 2015; Shen et al., 2023). However, a key distinction of our work is the multi-layered stakeholder process used to generate and test these scenarios. The landuse and landcover future scenarios that were developed with land-development stakeholders (Bremer et al., 2021; Brewington et al., 2020). Then these scenarios are tested using a simulation optimization framework that was co-designed by the modeling team and water governance stakeholders that are CWRM and BWS. This integration of two distinct, stakeholder-driven processes makes the analysis more credible, legitimate, and salient. It provides a scientifically credible and managerially salient answer to the policy question: What is the tangible 'groundwater value' of watershed restoration? Specifically, the process increased saliency by reporting metrics (e.g., spring discharge and submarine groundwater discharge) that regulators use. Furthermore, because the evaluation tool was co-designed with the end-users, the results carry a high degree of legitimacy, providing CWRM and BWS with a defensible basis to support investment in these synergistic strategies.

4.6 Societal preferences: Trade-off between spring discharge and pumping

This analysis quantifies the trade-off between maximizing groundwater withdrawal and preserving spring discharge, which is critical for local ecosystems and cultural practices (Burnett et al., 2020). Using the simulation-optimization framework, we evaluated sustainable yield under different spring flow constraints, defined as fractions of pre-development discharge. The results, shown in Fig. 10, reveal a clear inverse relationship: as the required spring discharge increases, the optimal sustainable withdrawal decreases. For instance, enforcing current spring discharge levels reduces the maximum potential withdrawal from 155 mgd to 121 mgd. The tool provides decision-makers with a clear, quantitative basis for navigating the competing demands of groundwater pumping and the

protection of ecosystem services including beneficial use and ecological and cultural resources of spring discharge.

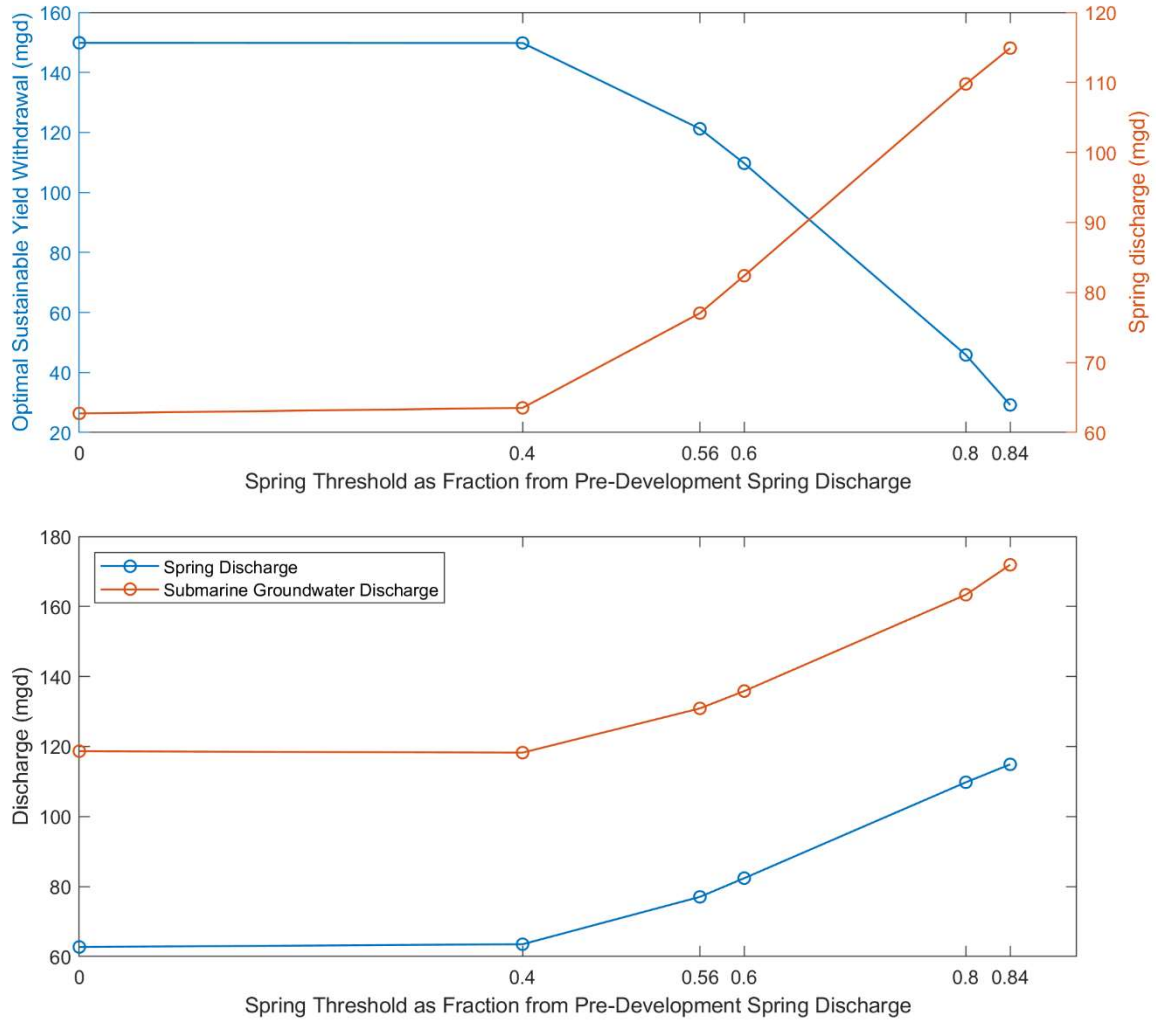


Fig. 10. Maximum allowable withdrawal, spring discharge and submarine groundwater discharge for 6 simulation optimization runs with different spring thresholds of 0, 40%, 56%, 60%, 80%, and 84%, respectively, from pre-development recharge. Threshold 0 refers to no spring discharge constraint. Threshold of 56% is the current spring discharge. Threshold of 84% is upper limit that the optimization can achieve.

This quantitative approach is transformed from a purely technical exercise into a decision-support tool through the collaborative modeling framework. The decision to frame the optimization around a values-based constraint that is spring discharge was a direct result of stakeholder engagement, making the model highly salient to their primary concerns. The resulting trade-off curve provides a legitimate basis for policy negotiation by transparently mapping the consequences of different choices, rather than prescribing a single optimal

solution. By incorporating stakeholder-defined constraints and providing clear, data-informed answers, the model's credibility is enhanced, allowing it to serve as an honest broker that grounds value-based discussions in quantitative reality.

Conclusions

This study introduces a stakeholder-informed simulation-optimization framework for evaluating and managing groundwater sustainability in coastal aquifers. The primary methodological contribution is the direct integration of collaborative modeling into the core formulation of the optimization problem, rather than treating stakeholder participation as a peripheral step. By coupling a three-dimensional, density-dependent groundwater model with an optimization algorithm and embedding stakeholder-defined objectives and constraints, the framework successfully operationalizes societal values and management realities within a scientifically rigorous tool. Moreover, the framework enables explicit evaluation of future scenarios, including those driven by climate change, land management, and socio-economic factors. It highlights the importance of incorporating uncertainty, intergenerational trade-offs, and nature-based solutions into groundwater governance. Therefore, the collaborative modeling approach for co-design of simulation optimization with stakeholders can be adapted to diverse hydrogeologic and institutional contexts, making it a valuable tool for advancing groundwater sustainability goals globally. The application to Hawai‘i’s Pearl Harbor aquifer demonstrates the framework's ability to navigate the intricate balance between maximizing water supply, mitigating salinization risk, and preserving ecological and cultural resources through maintaining ecological flow.

A key strength of the approach is the active engagement of stakeholders in co-design of a distributed modeling tool with calibrated numerical models, which enhances both model relevance and institutional acceptance. Stakeholder input informs the selection of decision variables, performance metrics, and scenario priorities, thereby aligning scientific outputs with regulatory mandates, community values, and operational constraints. In contrast to traditional analytical or lumped models, this framework accommodates spatial heterogeneity with embedded stakeholders’ knowledge, which is essential for sustainability assessments. Results demonstrate that relying solely on lumped models (e.g., RAM2) and oversimplified methods (e.g., a purely technical simulation optimization tool without co-design with stakeholders) may significantly overestimate sustainable groundwater yield and ignore management nuance, respectively. This can potentially undermine ecosystem services and long-term groundwater sustainability.

This work demonstrates the value of collaborative modeling and the co-design process. With the absence of this engagement, a purely technical simulation optimization could overstate potential gains, miss crucial institutional trade-offs, or yield solutions that are infeasible to

implement. By co-designing the scenarios and model structure with stakeholders, the framework becomes more credible, legitimate, and salient. The process generated salient output by directly addressing stakeholder questions, such as the relative impacts of different climate stressors and the tangible water value of watershed restoration. It enhanced legitimacy by providing a transparent and defensible basis for policy negotiations, particularly regarding sensitive issues like spring discharge protection. Finally, making modeling assumptions explicit and negotiated built trust and improved the credibility of the results for decision-makers.

The results provide several key insights for water managers and regulators. Projections show that under future climate scenarios, sustainable withdrawal from the aquifer could collapse by as much as 46% due to reduced recharge and sea-level rise. The work presents two mitigation tools that are pumping optimization and nature-based solutions. First, the application of stakeholder-informed pumping optimization can significantly mitigate these impacts, improving freshwater availability by up to 30% compared to non-optimized strategies. Second, the analysis quantifies the synergistic benefits of combining optimized pumping with nature-based solutions, such as watershed forest protection, which can add up to 10% (11 million gallons per day) to the sustainable yield. Furthermore, the framework provides a transparent tool for navigating critical policy trade-offs, such as the inverse relationship between groundwater withdrawal and the preservation of culturally and ecologically vital spring discharge.

In conclusion, this study offers a transferable and replicable process model for advancing groundwater sustainability in coastal regions worldwide. By operationalizing stakeholder knowledge within a powerful quantitative framework, this approach provides a more robust and equitable path for decision-making in data-limited, socially complex, and climate-stressed environments. It equips water managers, regulators, and researchers with a tool to better evaluate trade-offs and develop resilient management strategies, ultimately fostering more effective and sustainable groundwater governance.

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Data availability statement

The data that support this study is available at (, 2025).

Author contributions

Competing interests

None

Ethical statement

No human subjects are involved in this study.

AI assistance statement

ChatGPT-5 and Gemini 2.5 Pro were used to provided review comments, and to improve text clarity, succinctness, logical flow, and overall polish. AI contributions were verified for accuracy.

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